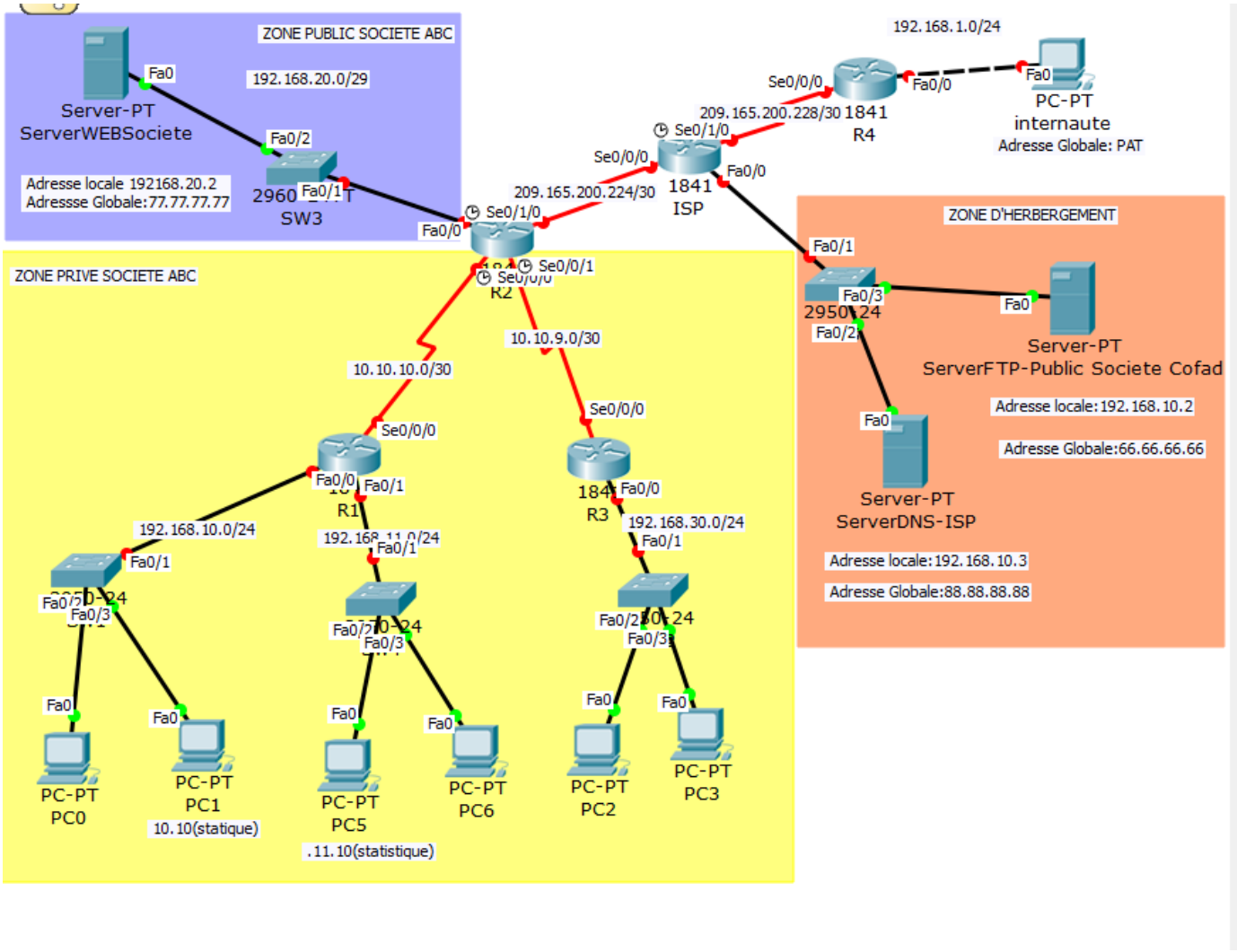
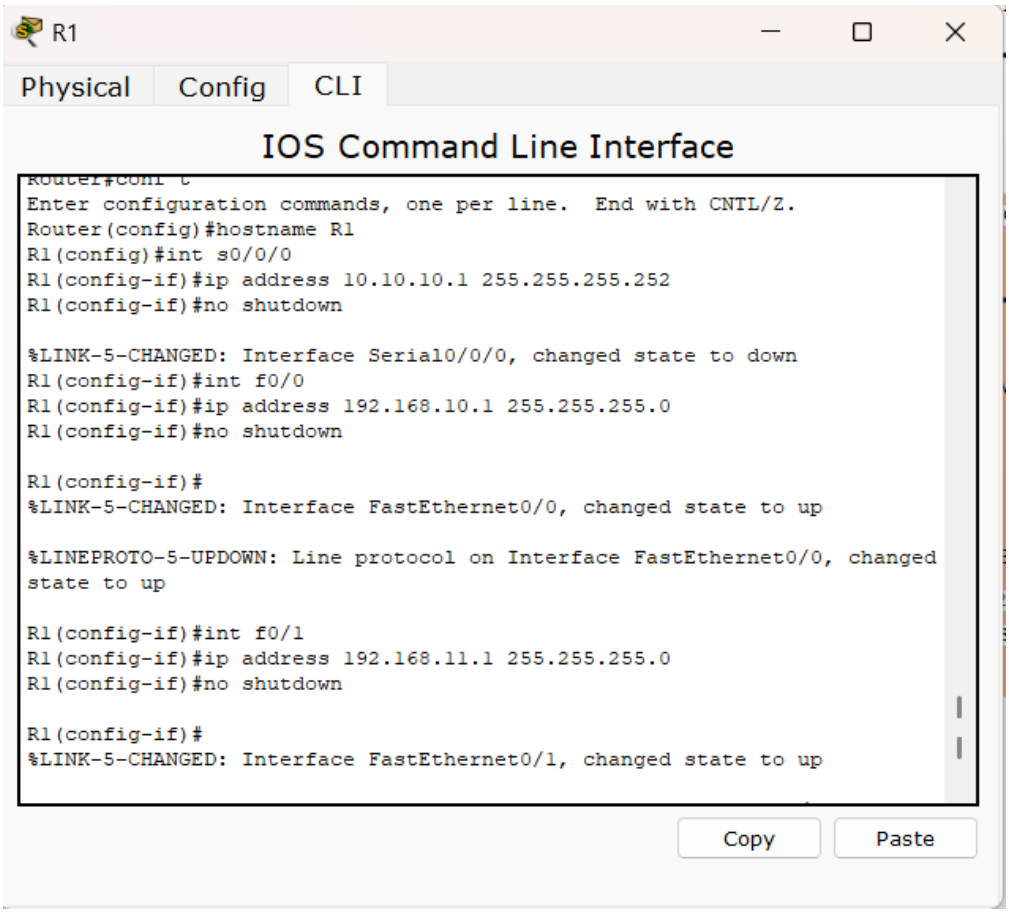
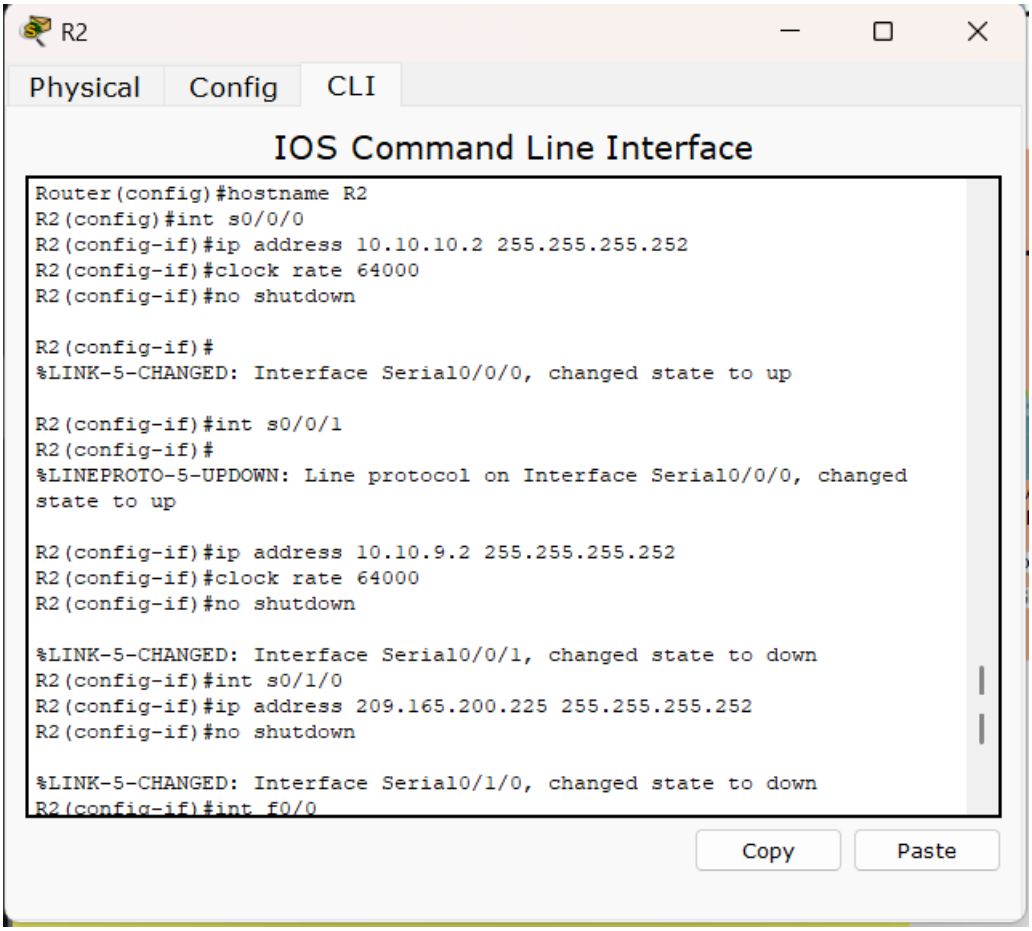
Compte-rendu : TP3 : NAT-PAT-DHCP

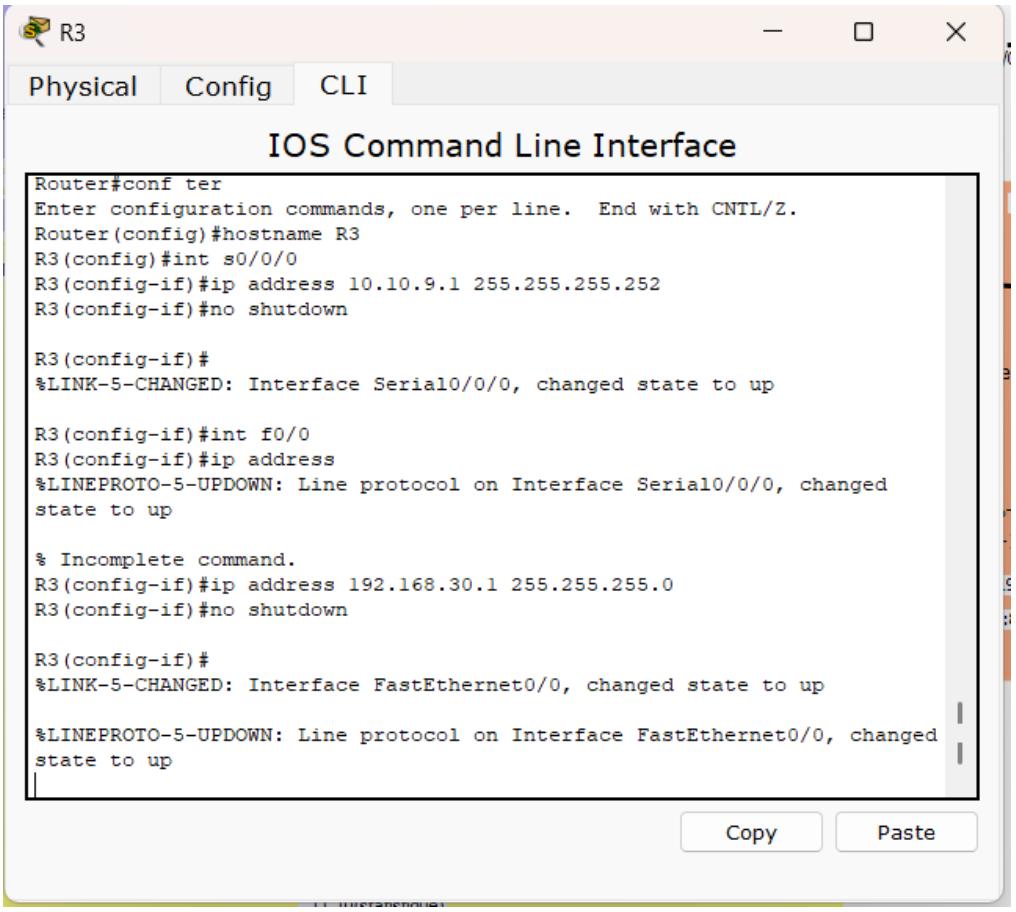
Malek Boughdiri RSI31

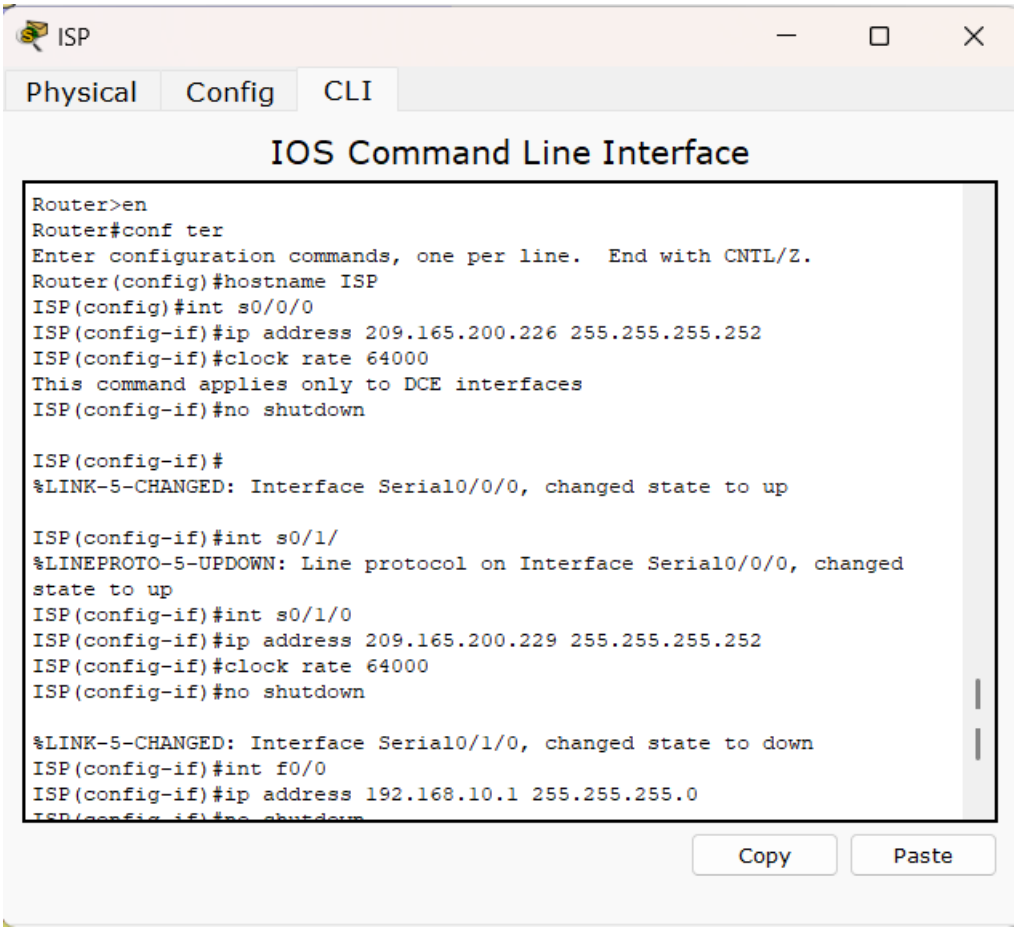
PARTIE A : Construire la topologie

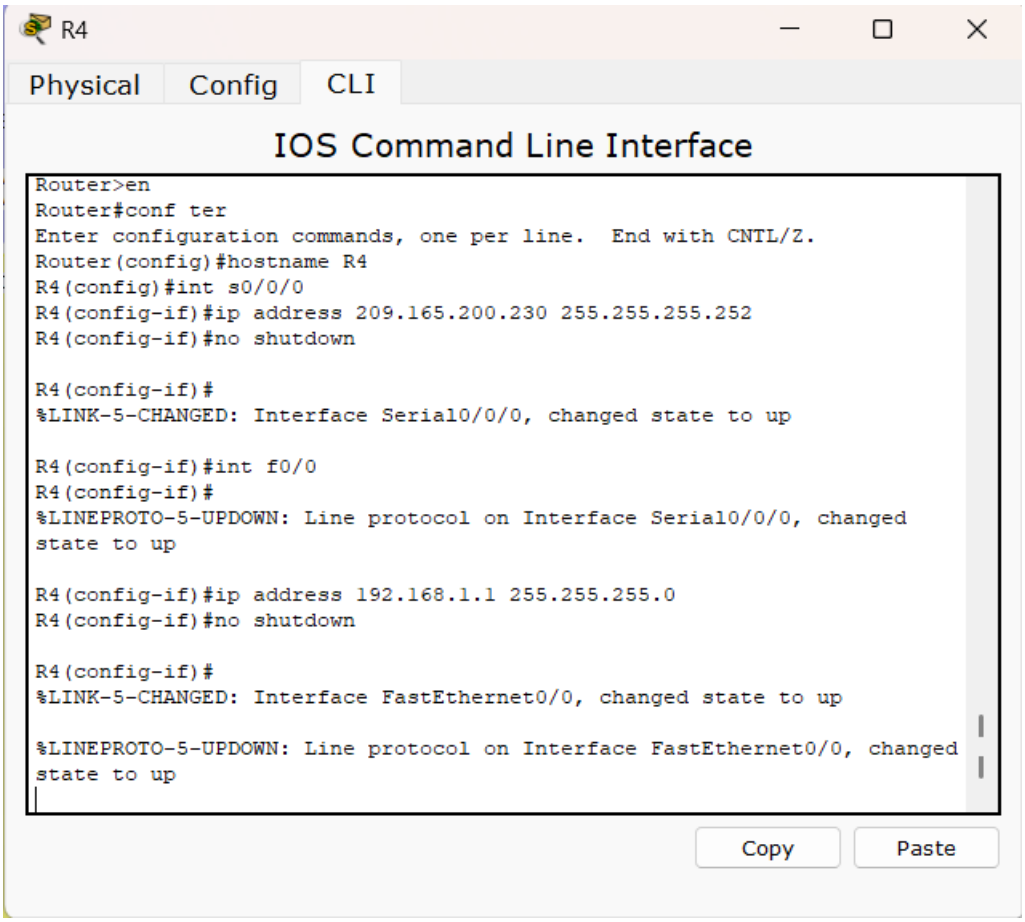



R1
Physical
Config
CLI
IOS Command Line Interface
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#int s0/0/0
R1(config-if)#ip address 10.10.10.1 255.255.255.252
R1(config-if)#no shutdown
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
R1(config-if)#int f0/0
R1(config-if)#ip address 192.168.10.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config-if)#int f0/1
R1(config-if)#ip address 192.168.11.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
Copy
Paste


R2
Physical
Config
CLI
IOS Command Line Interface
Router(config)#hostname R2
R2(config)#int s0/0/0
R2(config-if)#ip address 10.10.10.2 255.255.255.252
R2(config-if)#clock rate 64000
R2(config-if)#no shutdown
R2(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
R2(config-if)#int s0/0/1
R2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
R2(config-if)#ip address 10.10.9.2 255.255.255.252
R2(config-if)#clock rate 64000
R2(config-if)#no shutdown
%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down
R2(config-if)#int s0/1/0
R2(config-if)#ip address 209.165.200.225 255.255.255.252
R2(config-if)#no shutdown
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
R2(config-if)#int f0/0
Copy
Paste


R3
Physical
Config
CLI
IOS Command Line Interface
Router#conf ter
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R3
R3(config)#int s0/0/0
R3(config-if)#ip address 10.10.9.1 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
R3(config-if)#int f0/0
R3(config-if)#ip address
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed
state to up
% Incomplete command.
R3(config-if)#ip address 192.168.30.1 255.255.255.0
R3(config-if)#no shutdown
R3(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed
state to up
Copy
Paste


ISP
Physical
Config
CLI
IOS Command Line Interface
Router>en
Router#conf ter
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname ISP
ISP(config)#int s0/0/0
ISP(config-if)#ip address 209.165.200.226 255.255.255.252
ISP(config-if)#clock rate 64000
This command applies only to DCE interfaces
ISP(config-if)#no shutdown
ISP(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
ISP(config-if)#int s0/1/
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed
state to up
ISP(config-if)#int s0/1/0
ISP(config-if)#ip address 209.165.200.229 255.255.255.252
ISP(config-if)#clock rate 64000
ISP(config-if)#no shutdown
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
ISP(config-if)#int f0/0
ISP(config-if)#ip address 192.168.10.1 255.255.255.0
Copy
Paste



PARTIE B : Les tâches du TP

Tâche 1 : Routes statiques et routes par défaut

```
R1#conf ter
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip route 192.168.30.0 255.255.255.0 10.10.10.2
R1(config)#ip route 10.10.9.0 255.255.255.252 10.10.10.2
R1(config)#ip route 192.168.20.0 255.255.255.248 10.10.10.2
R1(config)#ip route 0.0.0.0 0.0.0.0 10.10.10.2
R1(config)#
```

```
R2(config)#ip route 192.168.10.0 255.255.255.0 10.10.10.1
R2(config)#ip route 192.168.11.0 255.255.255.0 10.10.10.1
R2(config)#ip route 192.168.30.0 255.255.255.0 10.10.9.1
R2(config)#ip route 0.0.0.0 0.0.0.0 209.165.200.226
R2(config)#
```

```
R3(config)#ip route 192.168.20.0 255.255.255.248 10.10.9.2
R3(config)#ip route 192.168.10.0 255.255.255.0 10.10.9.2
R3(config)#ip route 192.168.11.0 255.255.255.0 10.10.9.2
R3(config)#ip route 0.0.0.0 0.0.0.0 10.10.9.2
```

Test ping :

```
R1#ping 192.168.30.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.30.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 11/13/17 ms
```

Tâche 2 : Serveur DHCP sur R2

Étape 1 : adresses exclues (sur R2)

```
R2(config)#ip dhcp excluded-address 192.168.10.10
R2(config)#ip dhcp excluded-address 192.168.10.1
R2(config)#ip dhcp excluded-address 192.168.11.10
R2(config)#ip dhcp excluded-address 192.168.11.1
R2(config)#ip dhcp excluded-address 192.168.30.1
R2(config)#
```

Étape 2 : pools

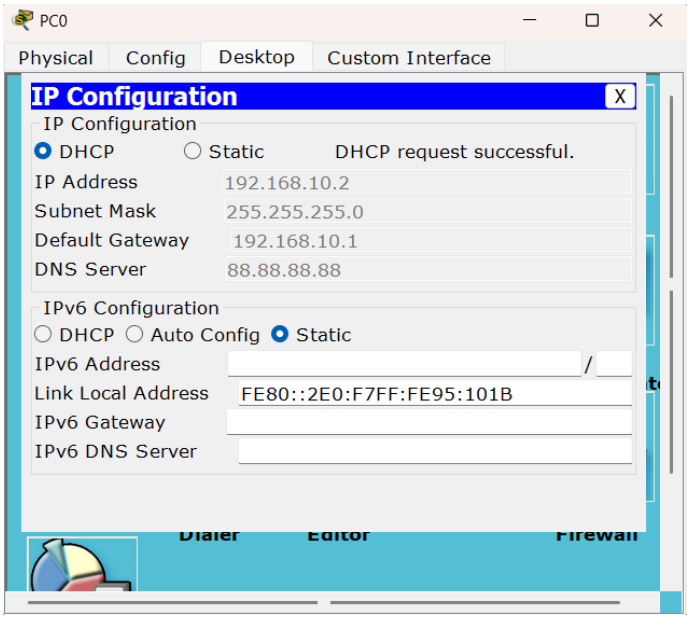
```
R2(config)#ip dhcp pool
R2(config)#ip dhcp pool R1f0/0
R2(dhcp-config)#network 192.168.10.0 255.255.255.0
R2(dhcp-config)#default-router 192.168.10.1
R2(dhcp-config)#dns-server 88.88.88.88
R2(dhcp-config)#exit
R2(config)#ip dhcp pool R1f0/1
R2(dhcp-config)#network 192.168.11.0 255.255.255.0
R2(dhcp-config)#default-router 192.168.11.1
R2(dhcp-config)#dns-server 88.88.88.88
R2(dhcp-config)#exit
R2(config)#ip dhcp pool R3f0/0
R2(dhcp-config)#network 192.168.30.0 255.255.255.0
R2(dhcp-config)#default-router 192.168.30.1
R2(dhcp-config)#dns-server 88.88.88.88
R2(dhcp-config)#exit
```

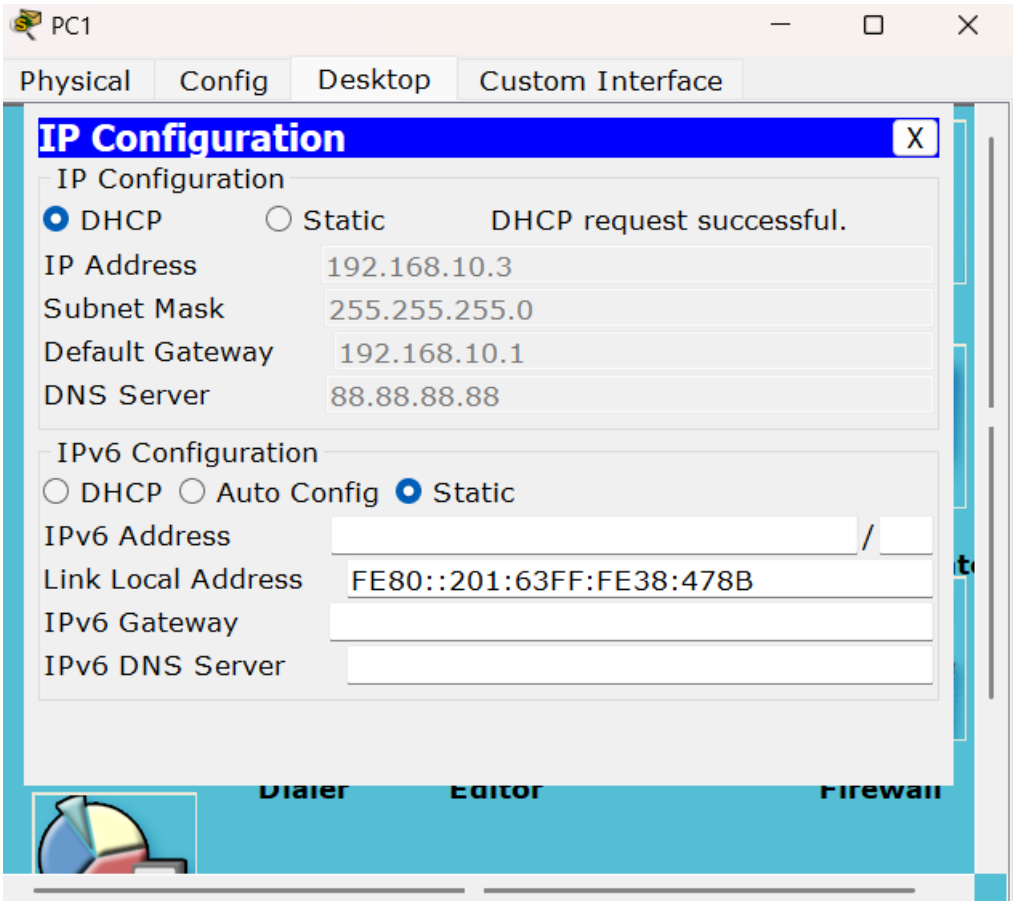
Étape 4 : agents relais

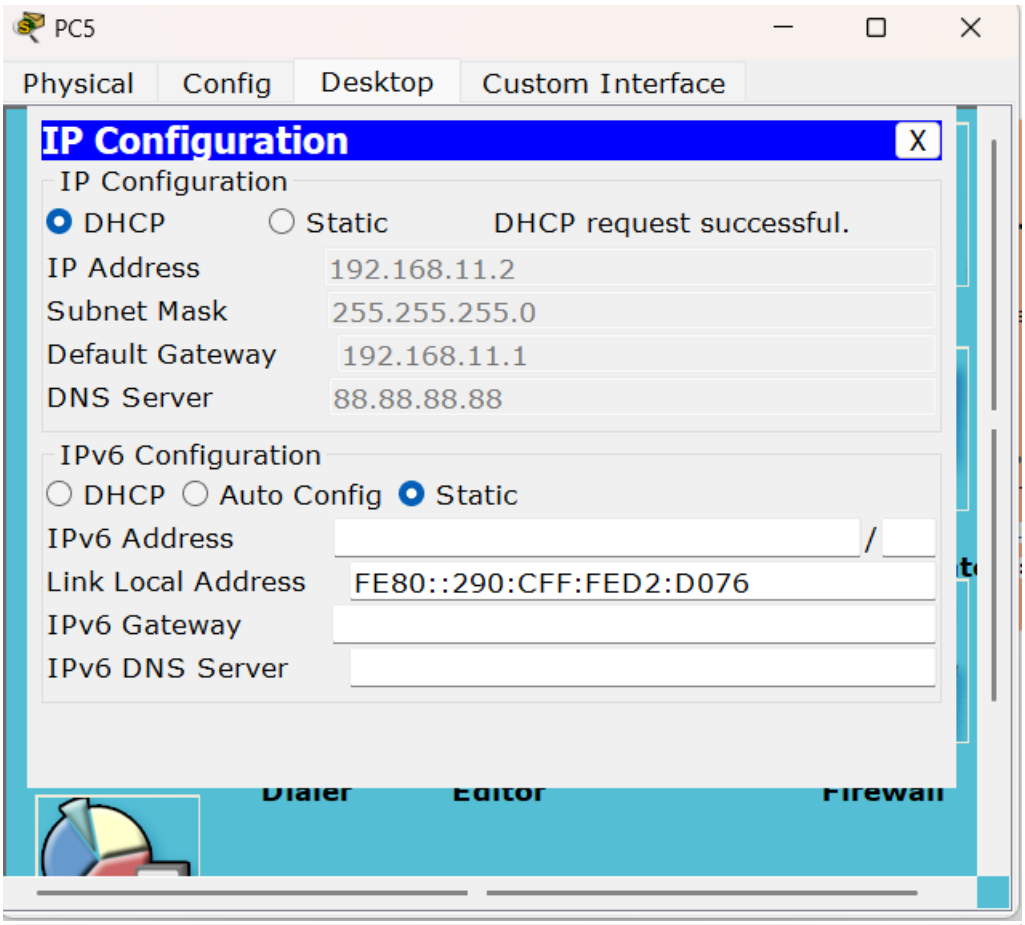
```
R1#conf ter
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#ip helper-address 10.10.10.2
R1(config-if)#int f0/1
R1(config-if)#ip helper-address 10.10.10.2
```

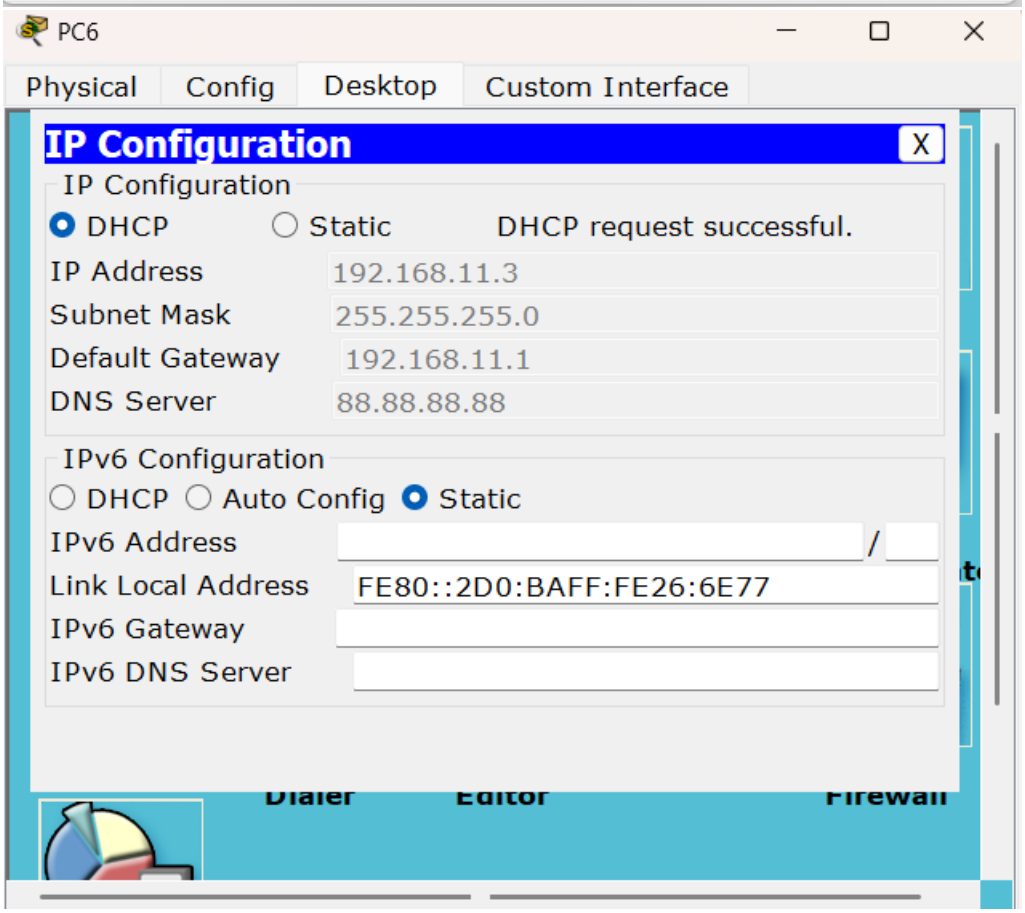
```
R3(config)#int f0/0
R3(config-if)#ip helper-address 10.10.9.2
```

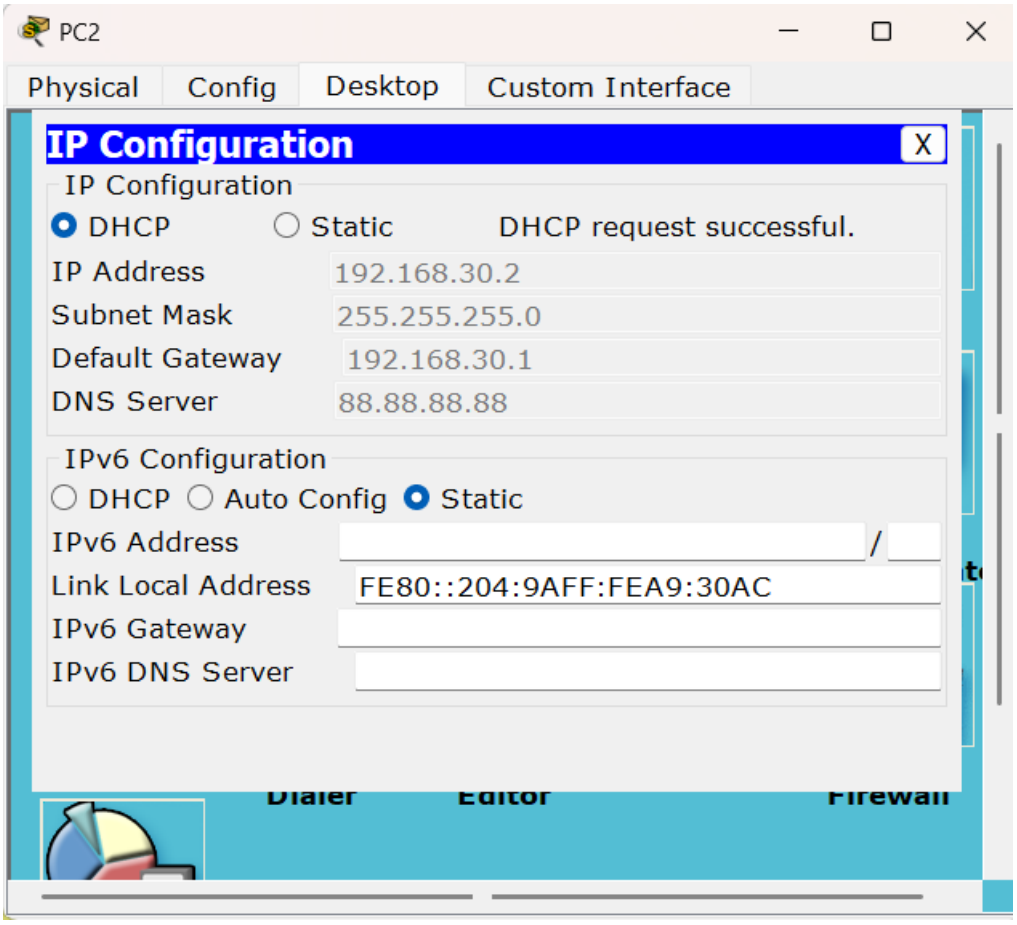
Vérification : sur PC1, PC6, PC2, PC3, onglet *Desktop* → *IP Configuration* → *DHCP*. Puis sur R2 :

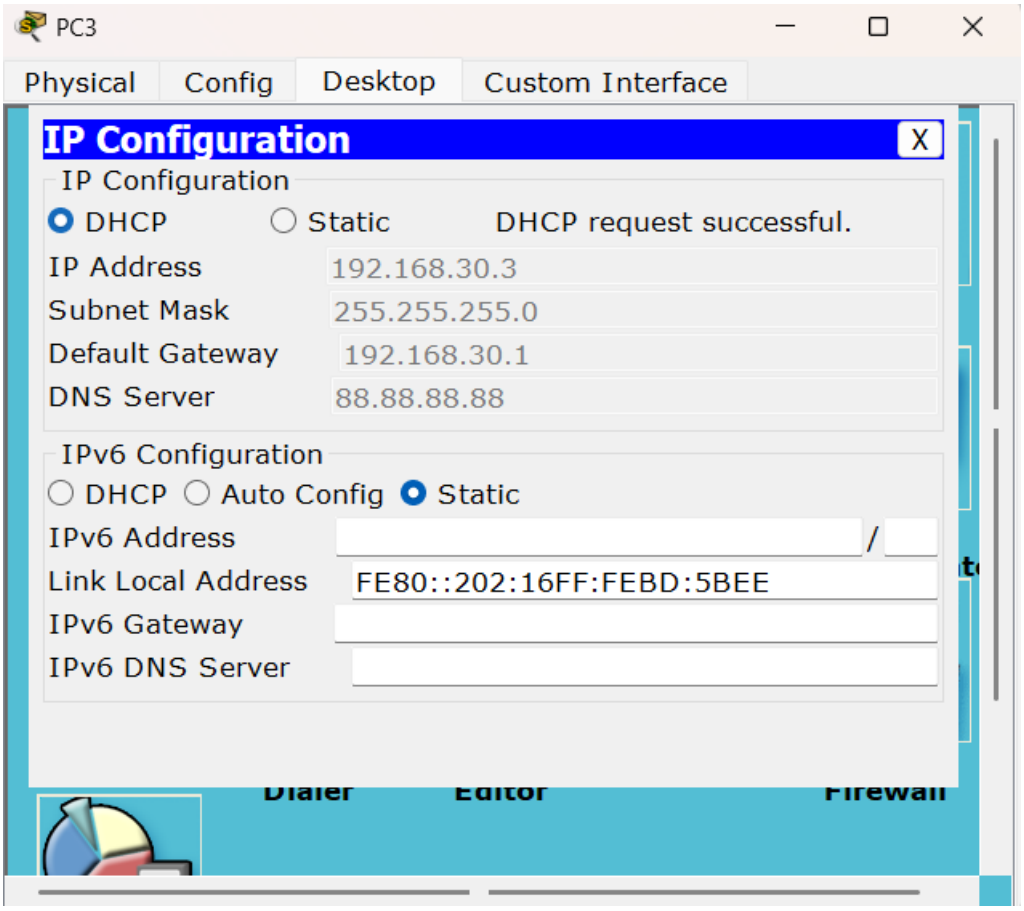





PC5
Physical
Config
Desktop
Custom Interface
IP Configuration
X
IP Configuration
DHCP
Static
DHCP request successful.
IP Address
192.168.11.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.11.1
DNS Server
88.88.88.88
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
/
Link Local Address
FE80::290:CFF:FED2:D076
IPv6 Gateway
IPv6 DNS Server
Dialer
Editor
Firewall


PC6
Physical
Config
Desktop
Custom Interface
IP Configuration
X
IP Configuration
DHCP
Static
DHCP request successful.
IP Address
192.168.11.3
Subnet Mask
255.255.255.0
Default Gateway
192.168.11.1
DNS Server
88.88.88.88
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
/
Link Local Address
FE80::2D0:BAFF:FE26:6E77
IPv6 Gateway
IPv6 DNS Server
Dialer
Editor
Firewall


PC2
Physical
Config
Desktop
Custom Interface
IP Configuration
X
IP Configuration
DHCP
Static
DHCP request successful.
IP Address
192.168.30.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.30.1
DNS Server
88.88.88.88
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
/
Link Local Address
FE80::204:9AFF:FEA9:30AC
IPv6 Gateway
IPv6 DNS Server
Dialer
Editor
Firewall


PC3
Physical
Config
Desktop
Custom Interface
IP Configuration
X
IP Configuration
DHCP
Static
DHCP request successful.
IP Address
192.168.30.3
Subnet Mask
255.255.255.0
Default Gateway
192.168.30.1
DNS Server
88.88.88.88
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
/
Link Local Address
FE80::202:16FF:FEBD:5BEE
IPv6 Gateway
IPv6 DNS Server
Dialer
Editor
Firewall

```
R2#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
192.168.10.2     00E0.F795.101B           --
Automatic
192.168.10.3     0001.6338.478B           --
Automatic
192.168.11.2     0090.0CD2.D076           --
Automatic
192.168.11.3     00D0.BA26.6E77           --
Automatic
192.168.30.2     0004.9AA9.30AC           --
Automatic
192.168.30.3     0002.16BD.5BEE           --
Automatic
```

Tâche 3 : NAT statique (étiquetée « Tâche 6 » dans le PDF)

Étape 1 : serveur web de la société (sur R2)

```
R2(config)#ip nat inside source static 192.168.20.2 77.77.77.77
```

Étape 2 : interfaces inside/outside (sur R2)

```
R2(config)#int s0/1/0
R2(config-if)#ip nat outside
R2(config-if)#int f0/0
R2(config-if)#ip nat inside
```

Mappage statique des serveurs de l'ISP

```
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ip nat inside source static 192.168.20.2 77.77.77.77
R2(config)#
R2(config)#int s0/1/0
R2(config-if)#ip nat outside
R2(config-if)#int f0/0
R2(config-if)#ip nat inside
R2(config-if)#
R2(config-if)#ip nat inside source static 192.168.10.3 88.88.88.88
R2(config)#ip nat inside
% Incomplete command.
R2(config)#int f0/0
R2(config-if)#ip nat inside source static 192.168.10.2 66.66.66.66
R2(config)#int f0/0
R2(config-if)#ip nat inside
R2(config-if)#int s0/0/0
R2(config-if)#ip nat outside
R2(config-if)#
```

Tâche 4 : Routage statique sur l'ISP (« Tâche 5 » dans le PDF)

```
ISP(config)#ip route 77.77.77.77 255.255.255.255 serial 0/0/0
```

Tâche 5 : Vérification du NAT statique

```
R2#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside
global
---  66.66.66.66       192.168.10.2       ---                ---
---  77.77.77.77       192.168.20.2       ---                ---
---  88.88.88.88       192.168.10.3       ---                ---
```

```
ISP#show ip nat translations
ISP#
ISP#
```

Tâche 6 : NAT dynamique avec pool (« Tâche 7 »)

Étape 1 : pool

```
R2(config)#ip nat pool MY-NAT-POOL 209.165.200.241 209.165.200.246
netmask 255.255.255.248
```

Étape 2 : ACL étendue

```
R2(config)#ip access-list extended NAT
R2(config-ext-nacl)#permit ip 192.168.10.0 0.0.0.255 any
R2(config-ext-nacl)#permit ip 192.168.11.0 0.0.0.255 any
R2(config-ext-nacl)#permit ip 192.168.30.0 0.0.0.255 any
R2(config-ext-nacl)#exit
```

Étape 3 : association pool et ACL

```
R2(config)#ip nat inside source list NAT pool MY-NAT-POOL
```

Étape 4 : interfaces

```
R2(config)#int s0/0/0
R2(config-if)#ip nat inside
R2(config-if)#int s0/0/1
R2(config-if)#ip nat inside
R2(config-if)#int s0/1/0
R2(config-if)#ip nat outside
R2(config-if)#
```

Étape 5 : route sur l'ISP

```
ISP(config)#ip route 209.165.200.240 255.255.255.248 s0/0/0
```

Étape 6 : vérification. Faites un ping de PC1 vers 209.165.200.226, ou un ping étendu depuis R1 :

```
PC>ping 209.165.200.226

Pinging 209.165.200.226 with 32 bytes of data:

Reply from 209.165.200.226: bytes=32 time=2ms TTL=253
Reply from 209.165.200.226: bytes=32 time=12ms TTL=253
Reply from 209.165.200.226: bytes=32 time=18ms TTL=253
Reply from 209.165.200.226: bytes=32 time=10ms TTL=253

Ping statistics for 209.165.200.226:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 18ms, Average = 10ms
```

```
R2#sh ip nat translations
Pro  Inside global     Inside local       Outside local      Outside
global
---  77.77.77.77       192.168.20.2       ---                ---
```

```
R2#sh ip nat statistics
Total translations: 1 (1 static, 0 dynamic, 0 extended)
Outside Interfaces: Serial0/1/0
Inside Interfaces: FastEthernet0/0 , Serial0/0/0 , Serial0/0/1
Hits: 4  Misses: 31
Expired translations: 21
Dynamic mappings:
-- Inside Source
access-list NAT pool MY-NAT-POOL refCount 0
 pool MY-NAT-POOL: netmask 255.255.255.248
       start 209.165.200.241 end 209.165.200.246
       type generic, total addresses 6 , allocated 0 (0%), misses 0
R2#
```

Tâche 7 : Surcharge NAT / PAT (« Tâche 8 »)

Étape 1 : supprimer le pool (sur R2)

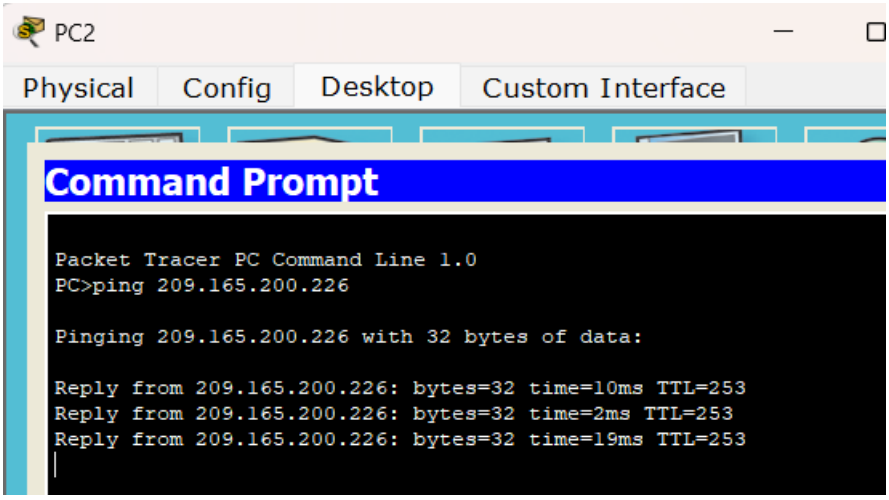
```
R2#clear ip nat translation *
R2#no ip nat inside source list NAT pool MY-NAT-POOL
      ^
% Invalid input detected at '^' marker.

R2#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#no ip nat inside source list NAT pool MY-NAT-POOL
R2(config)#no ip nat pool MY-NAT-POOL 209.165.200.241 209.165.200.246
netmask 255.255.255.248
```

Étape 2 : PAT

```
R2(config)#ip nat inside source list NAT interface s0/1/0 overload
```

Étape 3 : vérification. Faites des pings depuis PC1, PC2, PC6 vers 209.165.200.226, puis show ip nat translations : tous les hôtes apparaissent avec l'adresse 209.165.200.225 et des ports différents.




```
PC>ping 209.165.200.226

Pinging 209.165.200.226 with 32 bytes of data:

Reply from 209.165.200.226: bytes=32 time=16ms TTL=253
Reply from 209.165.200.226: bytes=32 time=13ms TTL=253
Reply from 209.165.200.226: bytes=32 time=21ms TTL=253
Reply from 209.165.200.226: bytes=32 time=10ms TTL=253

Ping statistics for 209.165.200.226:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 10ms, Maximum = 21ms, Average = 15ms
```

```
Packet Tracer PC Command Line 1.0
PC>ping 209.165.200.226

Pinging 209.165.200.226 with 32 bytes of data:

Reply from 209.165.200.226: bytes=32 time=2ms TTL=253
Reply from 209.165.200.226: bytes=32 time=20ms TTL=253
```

```
R2#sh ip nat translations
Pro  Inside global     Inside local       Outside local      Outside
global
icmp 209.165.200.225:1024192.168.11.3:1     209.165.200.226:1
209.165.200.226:1024
icmp 209.165.200.225:1025192.168.11.3:2     209.165.200.226:2
209.165.200.226:1025
icmp 209.165.200.225:1026192.168.11.3:3     209.165.200.226:3
209.165.200.226:1026
icmp 209.165.200.225:1027192.168.11.3:4     209.165.200.226:4
209.165.200.226:1027
icmp 209.165.200.225:1 192.168.30.2:1     209.165.200.226:1
209.165.200.226:1
icmp 209.165.200.225:2 192.168.30.2:2     209.165.200.226:2
209.165.200.226:2
icmp 209.165.200.225:3 192.168.30.2:3     209.165.200.226:3
209.165.200.226:3
icmp 209.165.200.225:4 192.168.30.2:4     209.165.200.226:4
209.165.200.226:4
---  77.77.77.77       192.168.20.2       ---                ---
```

Tâche 8 : PAT sur Router4 pour l'Internaute (« Tâche 9 »)

```
R4#conf ter
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#access-list 1 permit 192.168.1.0 0.0.0.255
R4(config)#ip nat inside source list 1 interface s0/0/0
R4(config)#ip nat inside source list 1 interface s0/0/0 overload
R4(config)#int f0/0
R4(config-if)#ip nat inside
R4(config-if)#int s0/0/0
R4(config-if)#ip nat outside
R4(config-if)#exit
R4(config)#ip route 0.0.0.0 0.0.0.0 209.165.200.229
R4(config)#
```

```
PC>ping 209.165.200.229

Pinging 209.165.200.229 with 32 bytes of data:

Reply from 209.165.200.229: bytes=32 time=2ms TTL=254
Reply from 209.165.200.229: bytes=32 time=1ms TTL=254
Reply from 209.165.200.229: bytes=32 time=1ms TTL=254
Reply from 209.165.200.229: bytes=32 time=12ms TTL=254

Ping statistics for 209.165.200.229:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 12ms, Average = 4ms
```